PROPER SPACES OVER QCQS BASES ARE LIMITS OF PROPER FINITE PRESENTATION ONES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3145-3182, following the statement at 3136-3143.

Editorial note. No omissions; the scheme-theoretic inverse image step is displayed.

Allowed standard prerequisites (editorial).

- Separated spaces are closed in finite presentation ones. proposition-separated-closed-in-finite-presentation (same chapter, cited at ~3146-3148). Closed immersion X -> X' with X' separated fp over Y.
- Closed subspaces are limits of closed fp ones. lemma-closed-is-limit-closed-and-finite-presentation (same chapter, cited at ~3149-3151). X = lim X_i with X_i -> X' closed immersion fp.
- Proper is etale-local on the base. Morphisms of Spaces, Lemma spaces-morphisms-lemma-proper-local (cross-chapter, cited at ~3155-3156). Reduces Y to an affine scheme.
- Weak Chow's lemma for algebraic spaces. Cohomology of Spaces, Lemma spaces-cohomology-lemma-weak-chow (cross-chapter, cited at ~3157-3159). Diagram with X" -> P^n_Y an immersion and pi : X" -> X' proper surjective.
- Properness and closed immersion permanence. Morphisms of Spaces, Lemmas spaces-morphisms-lemma-composition-proper, spaces-morphisms-lemma-universally-closed-permanence, spaces-morphisms-lemma-immersion-when-closed (cross-chapter, cited at ~3170-3175). pi^{-1}(X) -> P^n_Y is a closed immersion.
- Eventually closed immersion. lemma-eventually-closed-immersion (same chapter, cited at ~3176-3178). X_i' -> P^n_Y closed immersion for i large.
- Image of a proper morphism is proper. Morphisms of Spaces, Lemma spaces-morphisms-lemma-image-proper-is-proper (cross-chapter, cited at ~3180-3181). X_i -> Y proper.

Source reference. Original lemma 0A0U: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2984–2984. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0A0T: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1617–1617. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has

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to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 0873: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 3091–3091. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 089J: focused-seeds/geometry/S9-spaces-limits-proper-limit-of-proper-finite-presentation/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04XY: focused-seeds/geometry/S9-spaces-limits-proper-limit-of-proper-finite-presentation/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08AJ: focused-seeds/geometry/S9-spaces-limits-proper-limit-of-proper-finite-presentation/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04CD: focused-seeds/geometry/S9-spaces-limits-proper-limit-of-proper-finite-presentation/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 083R: focused-seeds/geometry/S9-spaces-limits-proper-limit-of-proper-finite-presentation/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NX: focused-seeds/geometry/S9-spaces-limits-proper-limit-of-proper-finite-presentation/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a proper morphism of algebraic spaces over S with Y quasi-compact and quasi-separated. Then $X = \lim X_i$ is a directed limit of algebraic spaces X_i proper and of finite presentation over Y and with transition morphisms and morphisms $X \to X_i$ closed immersions.*

Proof. By Proposition 0873 we can find a closed immersion $X \to X'$ with X' separated and of finite presentation over Y. By Lemma 0A0U we can write $X = \lim X_i$ with $X_i \to X'$ a closed immersion of finite presentation. We claim that for all i large enough the morphism $X_i \to Y$ is proper which finishes the proof.

To prove this we may assume that Y is an affine scheme, see Morphisms of Spaces, Lemma 083R. Next, we use the weak version of Chow's lemma, see Cohomology of Spaces, Lemma 089J, to find a diagram

$$\begin{array}{ccccc} X' & \xleftarrow{\pi} & X'' & \longrightarrow & \mathbf{P}^n_Y \\ & \searrow & \downarrow & \swarrow & \\ & & Y & & \end{array}$$

where $X'' \to \mathbf{P}^n_Y$ is an immersion, and $\pi : X'' \to X'$ is proper and surjective. Denote $X'_i \subset X''$, resp. $\pi^{-1}(X)$ the scheme theoretic inverse image of $X_i \subset X'$, resp. $X \subset X'$. Then $\lim X'_i = \pi^{-1}(X)$. Since $\pi^{-1}(X) \to Y$ is proper (Morphisms of Spaces, Lemmas 04XY), we see that $\pi^{-1}(X) \to \mathbf{P}^n_Y$ is a closed immersion (Morphisms of Spaces, Lemmas 04NX and 04CD). Hence for i large enough we find that $X'_i \to \mathbf{P}^n_Y$ is a closed immersion by Lemma 0A0T. Thus X'_i is proper over Y. For such i the morphism $X_i \to Y$ is proper by Morphisms of Spaces, Lemma 08AJ. $\square$